

## CSCE 637 COMPLEXITY THEORY · MODULE 07 · LECTURE NOTES

# Hierarchies and Barriers

*What more resources buy, and why the method stops working.*

---

**MODULE OUTCOMES**

- Prove the time hierarchy theorem.
- State the space hierarchy theorem and compare.
- Explain why hierarchy theorems do not separate P from NP.
- Explain relativisation and the Baker–Gill–Solovay result.
- Explain what a barrier result is and why it counts as progress.

## 1 The hierarchy theorems

---

CLAIM: there is a language decidable in time  $n^3$  and not in time  $n^2$ . More time buys strictly more power.

THE PROOF is CSCE 627 Module 05, with a clock added.

```
Define D(M):           # M is a machine encoding
    simulate M on input M for  $n^{2.5}$  steps
    if it halts and accepts, REJECT
    otherwise, ACCEPT
```

D runs in about  $n^3$  (the simulation overhead is logarithmic, which is why the exponents have room).

Suppose some machine  $M_0$  decides D's language in time  $n^2$ . Run D on  $M_0$ 's own encoding. The simulation COMPLETES, because  $n^2 < n^{2.5}$ . So D does the opposite of  $M_0$  on that input. But D's language is supposed to BE  $M_0$ 's language. CONTRADICTION.

SO:  $\text{TIME}(n^2)$  is strictly contained in  $\text{TIME}(n^3)$ .

THE TWO INGREDIENTS

```
DIAGONALISATION -- do the opposite of what the
    enumerated machine does (CSCE 627 Module 05)
A CLOCK -- the simulation must be cut off, and that
    cut-off is what makes this a COMPLEXITY result
    rather than a computability one
```

THE SPACE HIERARCHY is the same argument and is TIGHTER, because space simulation has no overhead at all -- so even a logarithmic factor more space buys strictly more power.

#### WHAT THE HIERARCHY THEOREMS GIVE, AND WHAT THEY DO NOT

**They give:**  $P \neq \text{EXPTIME}$ ,  $NL \neq \text{PSPACE}$ ,  $L \neq \text{PSPACE}$ , and a strict infinite hierarchy within each resource (Module 05 §2's landscape, where these are the only proved separations).

**So they do separate complexity classes**, which is more than the field manages anywhere else — and it is worth noting that the separations we have are all from this one technique.

**But only when the resource gap is large.** The simulation overhead means you need more than a constant factor, and more fundamentally **the technique works by comparing a resource bound against a strictly larger one of the same kind.**

**And P versus NP is not of that shape.** It compares deterministic and non-deterministic polynomial time — the same resource bound, differing only in the machine's mode — which diagonalisation has no handle on, because there is no larger bound to diagonalise into. §2 makes that intuition into a theorem.

## 2 Relativisation

**BAKER–GILL–SOLOVAY: ORACLES GO BOTH WAYS, SO DIAGONALISATION CANNOT SETTLE IT**

**There is an oracle  $A$  for which  $P^A = NP^A$ , and an oracle  $B$  for which  $P^B \neq NP^B$ . Both are constructed explicitly** —  $A$  can be taken to be any PSPACE-complete problem, and  $B$  is built by a diagonalisation.

**And diagonalisation arguments *relativise*** — they go through entirely unchanged when every machine in the argument is given the same oracle, because the argument never inspects what the machines do internally. **CSCE 627 Module 07 §1 observed exactly this for the halting proof.**

**So no relativising proof can settle  $P$  versus  $NP$ :** any such proof would establish the same answer relative to every oracle, and the answer demonstrably differs between  $A$  and  $B$ . **The technique is not merely insufficient; it is provably incapable.**

**Which rules out the entire technique that produced every separation in §1. That is a precise, proved statement about the inadequacy of the field's main tool, and it is from 1975** — four years after the question was posed, which is part of why the field's attitude to it is as it is.

| Barrier                                      | What it rules out                                                                                                                                                        | Year        |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Relativisation</b> (Baker, Gill, Solovay) | <b>Any proof technique that works unchanged in the presence of an oracle — which includes diagonalisation and simulation.</b>                                            | <b>1975</b> |
| <b>Natural proofs</b> (Razborov, Rudich)     | <b>Combinatorial circuit lower bound arguments of the kind that had been succeeding on restricted models (Module 09 §3), assuming strong one-way functions exist.</b>    | <b>1994</b> |
| <b>Algebrisation</b> (Aaronson, Wigderson)   | <b>Algebraic extensions of relativising techniques, including the arithmetisation method that had evaded the first barrier and proved <math>IP = PSPACE</math> (§3).</b> | <b>2008</b> |

**Each barrier followed a period of optimism about a technique — and the third was constructed specifically to determine whether arithmetisation, which had evaded the first barrier, could reach  $P$  versus  $NP$ . The history is the point:** each barrier closed off a route that serious people were pursuing, which is what makes them contributions rather than discouragements.

### 3 What gets past the first barrier

#### SOME RESULTS DO NOT RELATIVISE, WHICH IS WHY THERE IS HOPE

**IP = PSPACE** (Module 10 §2) **does not relativise** — there are oracles relative to which it is false. **So non-relativising techniques exist, and they have proved real and substantial theorems.** This was the news of the early 1990s.

**The technique is arithmetisation:** convert a Boolean formula into a polynomial over a finite field, and then exploit properties of low-degree polynomials — that they are determined by few points, that two distinct ones disagree almost everywhere — **which have no oracle analogue, because an oracle is an arbitrary function and polynomials are not.**

**Which is precisely why the algebrisation barrier was constructed** — to determine whether arithmetisation alone could reach P versus NP. **It cannot**, and the proof constructs algebraic oracles going both ways, in the same shape as Baker–Gill–Solovay.

**So the state of play is: three barriers, each closing a technique, and the open question is what a technique evading all three would even look like. Nobody knows, and that is the honest summary** — geometric complexity theory is the most developed proposal and has produced no separation.

### 4 Barriers as progress

- **It redirects effort.** After 1975, no serious researcher attempted a relativising proof of  $P \neq NP$  — which saved a great deal of time and is exactly CSCE 627 Module 06 §4's argument about proving impossibility early, one level up.
- **It is a theorem about proofs**, which is an unusual and respectable kind of result — **and it is CSCE 627's undecidability argument applied to a methodology** rather than to a problem.
- **It specifies what a solution must look like:** non-relativising, non-naturalising, and non-algebrising. **Three constraints is considerably more guidance than none**, and any claimed proof can be checked against them immediately — which is how most claimed proofs are dismissed.
- **And each barrier was proved by constructing an explicit counterexample**, which required genuine technique and produced other results along the way — the Baker–Gill–Solovay construction is a useful diagonalisation in its own right.
- **Which is the field's own honest assessment of its position: fifty years of not proving the central result, and three theorems explaining why the available methods cannot.** Few fields can point to proofs that their own methods are inadequate, and **treating that as intellectual honesty rather than as failure is the right reading.**

**WHAT TO TAKE FROM THIS MODULE**

**More resources provably help, when the gap is large. That is the positive result**, and it is proved by CSCE 627 Module 05's technique plus a clock — which is a satisfying reuse.

**And the same technique provably cannot settle P versus NP**, which is a **sharper and more useful statement than 'nobody has managed it'**.

**Two further barriers close two further techniques** (Module 09 §3's natural proofs, and the 2008 algebrisation result) — so the closure is not a single historical accident.

**So the honest position is: the question is open, the belief is empirical** (Module 03 §3), **and the field has mapped out three dead ends rigorously. Which is a form of progress, and it is what is available** — and a course that presented the consensus without the barriers would be giving a false impression of how much is known.

## Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

| Resource                                                                              | What it covers for this module                                                                                     |
|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Arora & Barak — chapters 3 and 23 (free draft)                                        | <b>The hierarchy theorems and all three barriers</b> — chapter 23 is the best treatment of the barriers available. |
| Baker, Gill & Solovay — Relativizations of the $P \stackrel{?}{=} NP$ Question (free) | <b>§2's result</b> , and both oracle constructions.                                                                |
| Aaronson & Wigderson — Algebrization: A New Barrier in Complexity Theory (free)       | <b>§2's third barrier</b> , with a good account of why it was needed.                                              |
| Aaronson — $P \stackrel{?}{=} NP$ , the barriers section (free)                       | <b>All three barriers assessed together</b> , with a candid view of what remains.                                  |

## Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Prove the time hierarchy theorem** in your own words, and identify where the clock enters.
- 2 **Explain why the space hierarchy is tighter.**
- 3 **Work out which separations in Module 05's landscape come from hierarchy theorems** and which are unproved.
- 4 **Explain why diagonalisation relativises**, by checking the hierarchy proof step by step with oracles added.

- 5 **State Baker–Gill–Solovay** and explain why it rules out a technique rather than an answer.
- 6 **Verify that a PSPACE-complete oracle makes  $P = NP$  relative to it.**
- 7 **Read about arithmetisation** and say why it does not relativise.
- 8 **Find a claimed proof of  $P \neq NP$  online** and check it against the three barriers.
- 9 **Write an argument for why barrier results count as progress**, and the strongest objection to it.
- 10 **State what a successful technique would have to avoid.**

## Self-check

---

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Prove the time hierarchy theorem and name the two ingredients.
- 2 Why is the space hierarchy tighter?
- 3 What do the hierarchy theorems separate, and what can they not?
- 4 Why is  $P$  versus  $NP$  not of the right shape for them?
- 5 State Baker–Gill–Solovay and what it rules out.
- 6 Why does diagonalisation relativise?
- 7 Name the three barriers and what each closes.
- 8 What technique gets past the first, and what did it prove?
- 9 Give five reasons a barrier result is a contribution.
- 10 State the honest position on  $P$  versus  $NP$ .